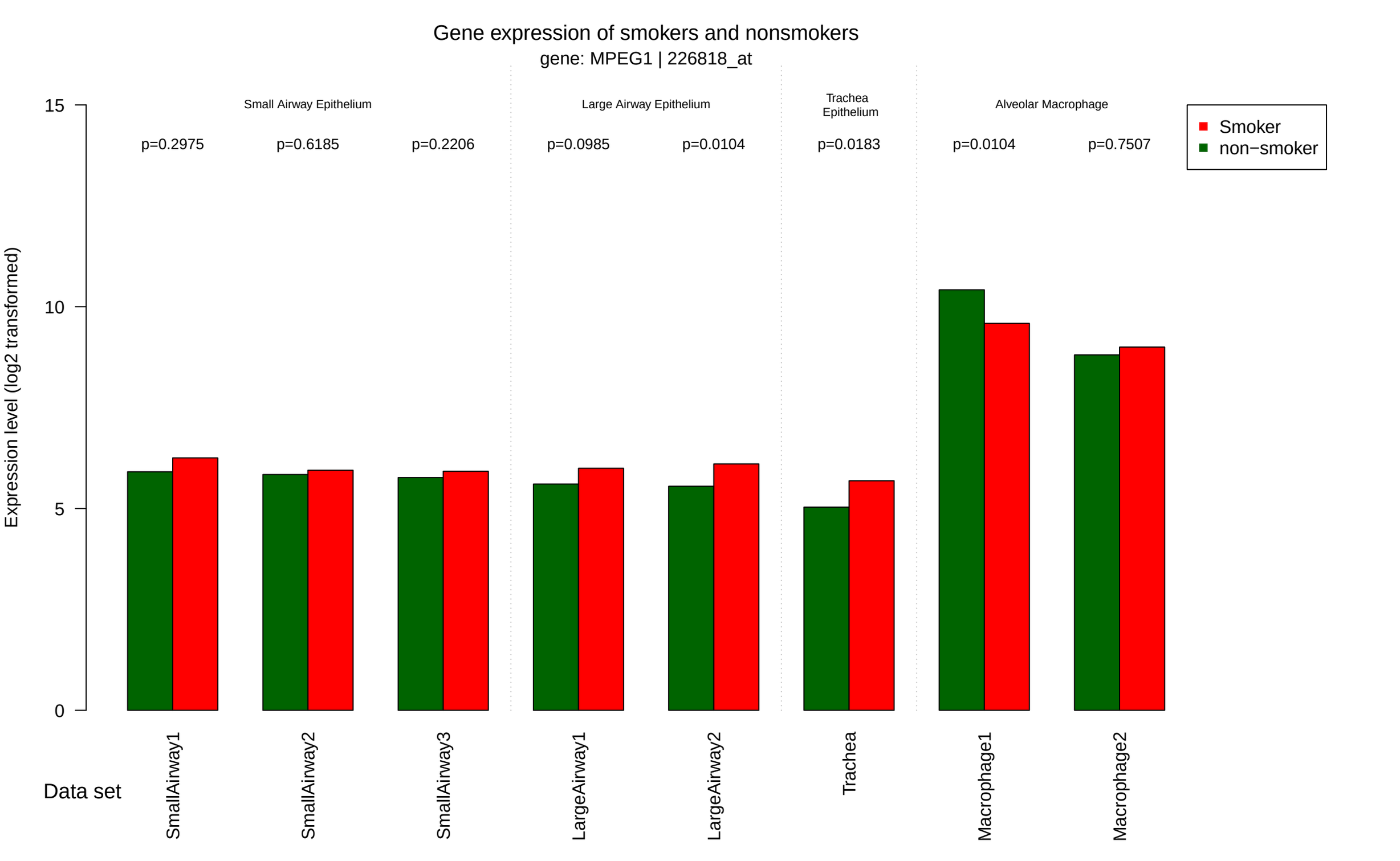
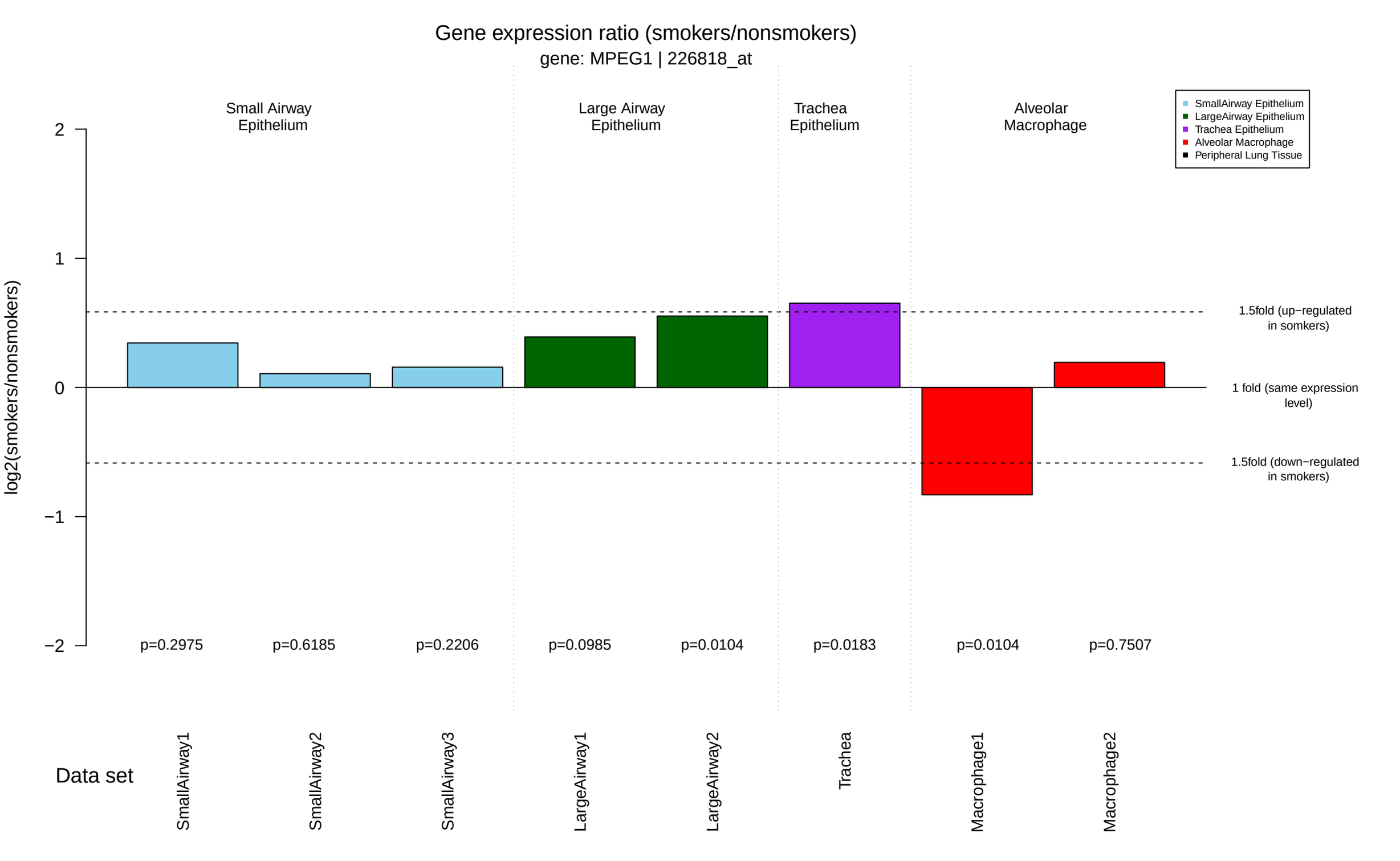
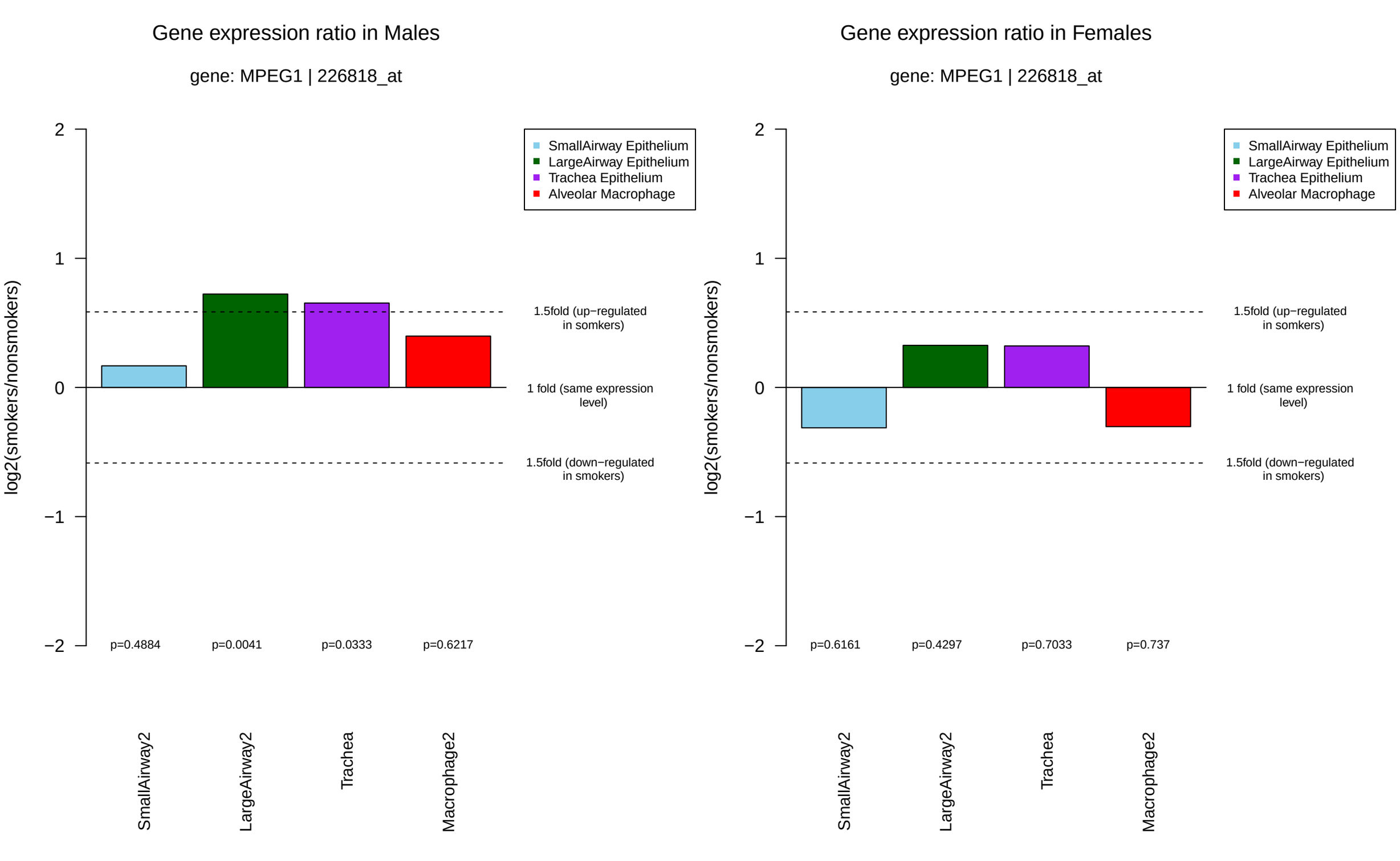
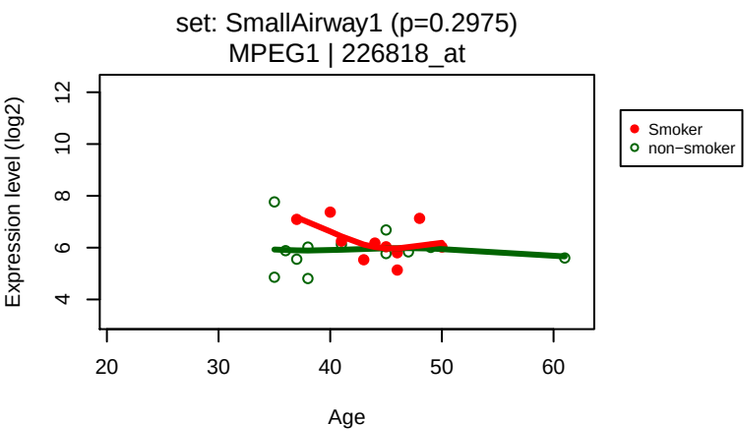
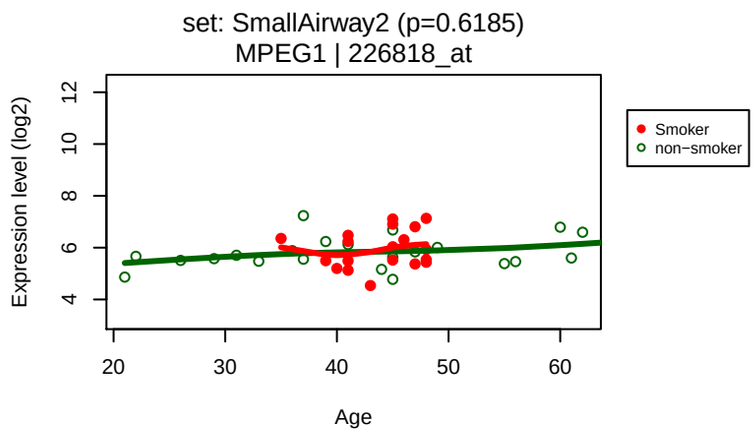
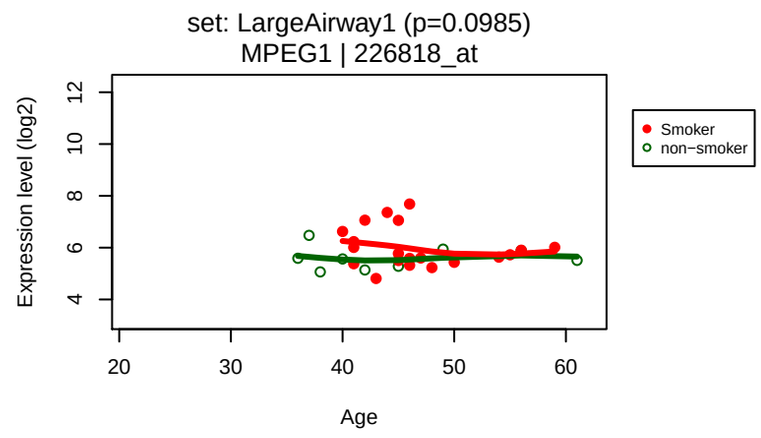

Gene expression of smokers and nonsmokers
gene: MPEG1 | 226818_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2975
p=0.6185
p=0.2206
p=0.0985
p=0.0104
p=0.0183
p=0.0104
p=0.7507
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

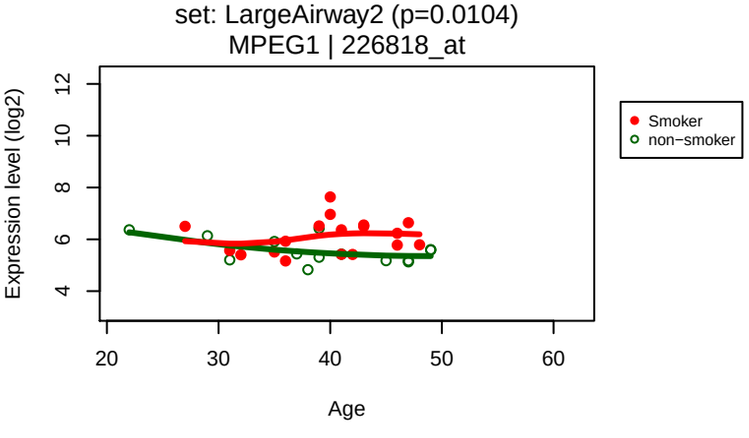

Gene expression ratio (smokers/nonsmokers)
gene: MPEG1 | 226818_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2975
p=0.6185
p=0.2206
p=0.0985
p=0.0104
p=0.0183
p=0.0104
p=0.7507
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

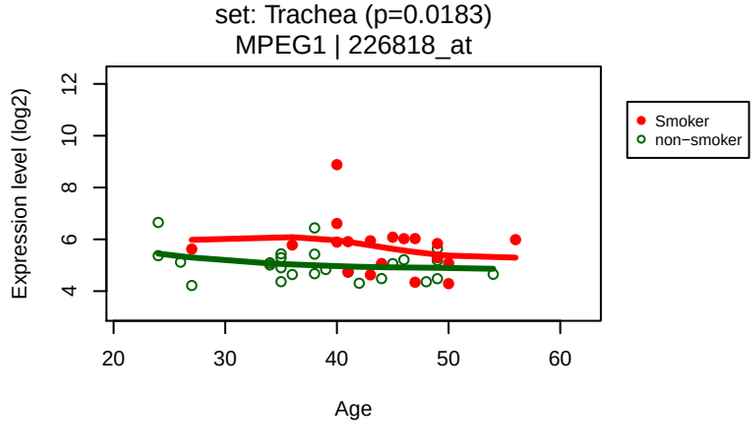

Gene expression ratio in Males
gene: MPEG1 | 226818_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4884
p=0.0041
p=0.0333
p=0.6217
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MPEG1 | 226818_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6161
p=0.4297
p=0.7033
p=0.737
SmallAirway2
LargeAirway2
Trachea
Macrophage2

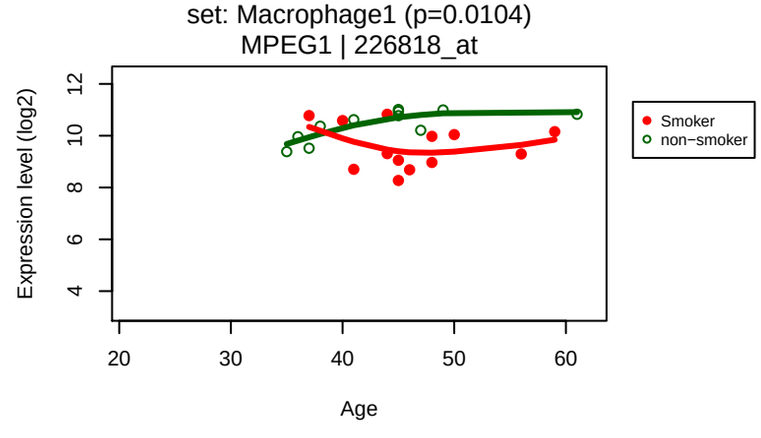

set: SmallAirway1 (p=0.2975)
MPEG1 | 226818_at
Expression level (log2)
Age
Smoker
non−smoker

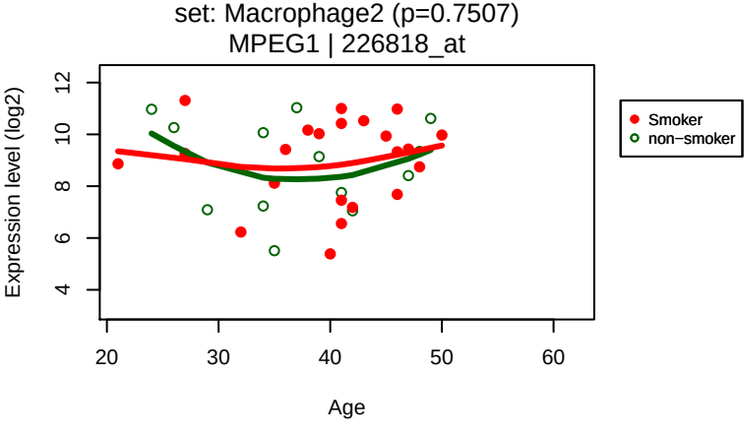

set: SmallAirway2 (p=0.6185)
MPEG1 | 226818_at
Expression level (log2)
Age
Smoker
non−smoker

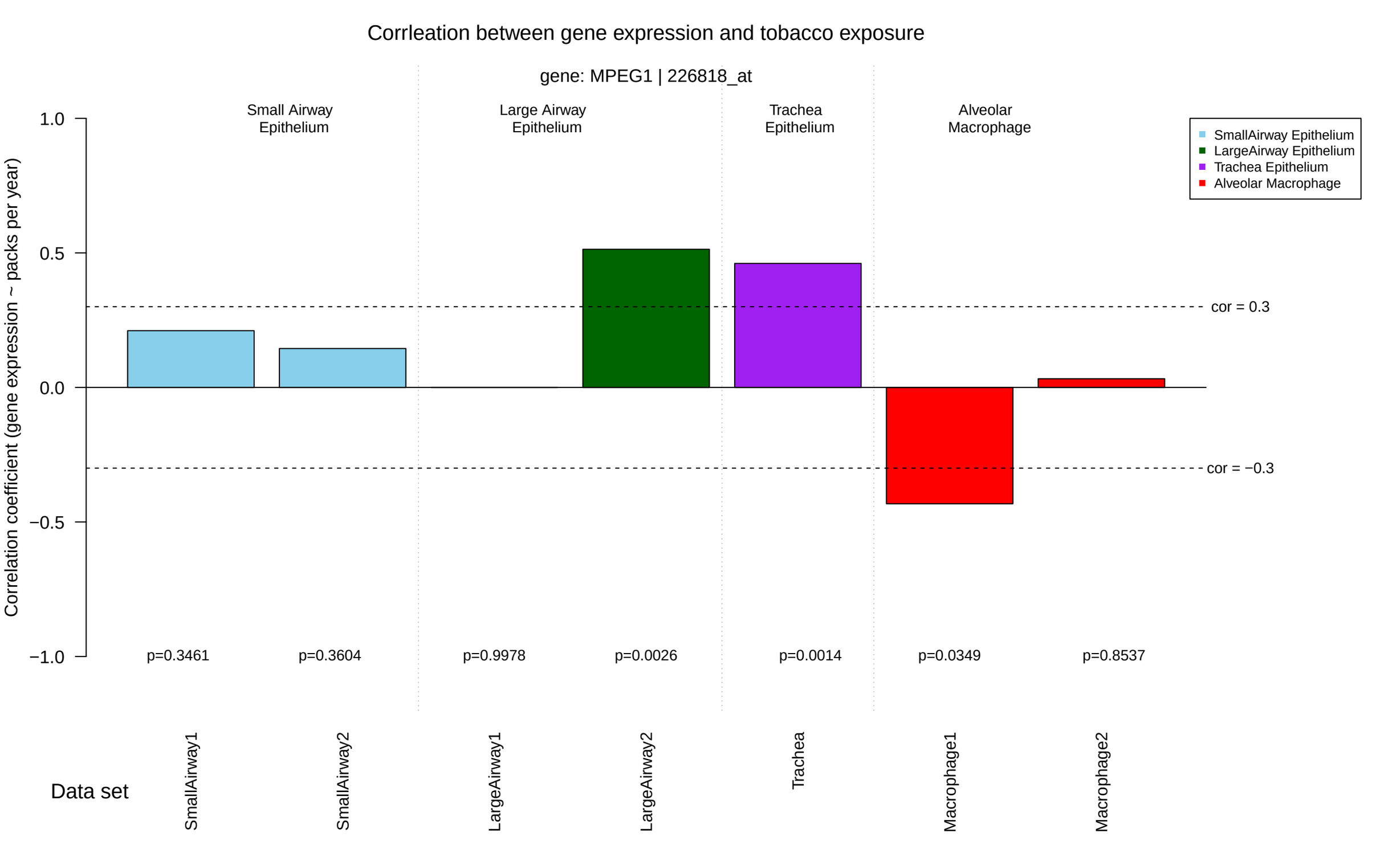

set: LargeAirway1 (p=0.0985)
MPEG1 | 226818_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0104)
MPEG1 | 226818_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0183)
MPEG1 | 226818_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0104)
MPEG1 | 226818_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7507)
MPEG1 | 226818_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MPEG1 | 226818_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3461
p=0.3604
p=0.9978
p=0.0026
p=0.0014
p=0.0349
p=0.8537
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2